

Human Activity Recognition Using Hidden Markov Models

Machine Learning Techniques II

El Hadji Faly Seck

Git repository: [falyseck/activity-recognition](https://github.com/falyseck/activity-recognition)

1. Background and Motivation

Human Activity Recognition (HAR) enables physical activity inference from wearable sensors for healthcare, smart homes, and fitness monitoring. This project develops a low-cost HAR system using iPhone accelerometer and gyroscope data to distinguish four activities (Still, Standing, Walking, Jumping). Hidden Markov Models (HMMs) are employed to model temporal dependencies between hidden activity states and observable sensor features, making them suitable for noisy time-series data.

2. Data Collection and Preprocessing

2.1 Recording Protocol

Data was collected using Sensor Logger on an iPhone (iOS), exported as separate CSV files per sensor with columns: time, seconds_elapsed, z, y, x. Five training recordings per activity were made in Session 1 (~6-10 seconds each at waist level), plus one unseen test recording per activity in Session 2. Raw data at ~100 Hz was resampled to 50 Hz.

Activity	Train Recordings	Train Samples	Test Samples
Still	5	1,927	338
Standing	5	1,893	370
Waling	5	1,868	413
Jumping	5	1,950	385

2.2 Preprocessing Pipeline

Raw data underwent: (1) File pairing by matching accelerometer and gyroscope CSVs by filename index; (2) Temporal alignment via `pandas.merge_asof()` for nearest-timestamp matching; (3) Resampling to 50 Hz using `scipy.signal.resample()`; (4) Axis renaming (ax, ay, az for accelerometer; gx, gy, gz for gyroscope); (5) Z-score normalisation using `sklearn.preprocessing.StandardScaler` fitted on training data only to prevent test set leakage.

3. Feature Extraction

A sliding window approach (50 samples = 1 second, 50% overlap) extracted 79 features per window:

- **Time-domain (60 features)**: Mean, standard deviation, variance, min, max, range, RMS, skewness for each of six sensor axes
- **Signal-level (4 features)**: Signal Magnitude Area (SMA), mean/standard deviation/maximum of resultant acceleration magnitude

- **Frequency-domain (12 features)**: Dominant frequency and spectral energy (Welch's method) for each sensor axis
- **Cross-axis correlations (3 features)**: Pearson correlations between ax-ay, ax-az, ay-az

Training feature matrix: 276 windows × 79 features; Test feature matrix: 54 windows × 79 features

4. HMM Implementation

4.1 Model Architecture

A one-HMM-per-class strategy was adopted, with separate Gaussian HMMs trained on each activity's feature sequences. Inference selects the class with highest log-likelihood for the test sequence.

Component	Configuration
Hidden States	4 internal states per model
Emission	Diagonal-covariance Gaussian
Classes	4 (Still, Standing, Walking, Jumping)
Library	Hmmlearn v0.3+

The four hidden states represent sub-phases of each activity (e.g., gait cycle phases for walking: weight transfer, push-off, swing, heel strike).

4.2 Training: Baum-Welch Algorithm

Baum-Welch (EM) ran for 200 iterations per model. Trained model log-likelihoods reflected activity complexity:

Activity	Log-Likelihood
Still	10,175.59
Standing	8,315.50
Walking	4,611.58
Jumping	-4,867.46

4.3 Decoding: Viterbi Algorithm

For each test sequence, Viterbi decoding finds the most probable hidden state path for each class HMM. The class with highest total log-probability is the predicted activity.

5. Results

5.1 Evaluation Setup

Test set: 4 new recordings (one per activity) from a separate session (chronological train/test split). Total test windows: 54 (12 Still, 13 Standing, 15 Walking, 14 Jumping).

5.2 Quantitative Results

Overall accuracy: **72.2%**. Still and Standing achieved perfect classification (sensitivity=1.0, specificity=1.0). Walking: all 15 windows misclassified as Jumping (sensitivity=0.0). Jumping: perfectly classified.

5.3 Visualisation Findings

- **Raw_signals.png**: az axis (vertical) clearly differentiates activities (~9.81 m/s² for Still/Standing, sinusoidal for Walking, high-amplitude impulses for Jumping)
 - **Transition_matrices.png**: Strong diagonal dominance (0.8-0.99) reflects temporal persistence of activity sub-phases
 - **Decoded_sequence.png**: Ground-truth vs Viterbi-decoded labels; Still, Standing, Jumping correct; Walking misclassified as Jumping
 - **Confusion_matrix.png**: Walking→Jumping is the sole error mode
 - **Feature_distributions.png**: Clean separability for Still/Standing but overlap between Walking/Jumping in acceleration features
- NB**: All figures can be found in the notebook. They were not included in the report for structure adherence.

5.4 Discussion

Walking misclassification stems from overlapping feature distributions in the 79-dimensional space, particularly in az-axis dominant frequency (~1.5-2.0 Hz for both) and spectral energy. The Jumping HMM learned a broader emission distribution from training data covering varied jump styles (including controlled hops resembling brisk walking), which "captured" walking test sequences.

6. Analysis and Reflection

6.1 Activity Discrimination Difficulty

- **Easiest:** Still vs Jumping (opposite feature space extremes: near-zero variance vs high-amplitude impulses)
- **Runner-up:** Standing vs Still (SMA and az variance capture subtle postural sway in Standing absent in Still)
- **Hardest:** Walking vs Jumping (both periodic at ~1.5-2.0 Hz with significant az variance)

6.2 Transition Probabilities

All HMMs showed strong self-transitions (0.8-0.99), accurately reflecting human activity persistence. The Jumping HMM had slightly lower self-transition probabilities, capturing the natural sub-phase cycle (airborne→landing→ground contact→push-off).

6.3 Sensor Noise and Sampling Rate

Resampling from ~100 Hz to 50 Hz is appropriate (Nyquist limit well above human movement frequencies). Temporal alignment via merge_asof introduced ~5-10 ms timing offsets between sensors, potentially adding spectral noise. Hardware-synchronised sensors would avoid this.

6.4 Improvement Opportunities

- **More training data:** 20+ recordings per activity covering varied styles would improve emission model coverage
- **Better Walking-Jumping discrimination:** Step detection (regular inter-peak intervals), barometric pressure (altitude change during jumps), peak-to-valley ratio on az (sharp vs smooth signals)
- **Unified multi-class HMM:** Single HMM with 4 explicit activity states would naturally model inter-activity transitions
- **Alternative classifiers:** LSTM or CNN baselines for comparison
- **More test recordings:** 3-5 per class for reliable accuracy estimates

7. Conclusion

A complete HAR pipeline was implemented using HMMs on iPhone sensor data: data collection via Sensor Logger, temporal alignment/resampling of separate CSV files, 79-dimensional feature extraction, Gaussian HMM training via Baum-Welch, and Viterbi decoding for prediction. The system achieves **72.2% accuracy** on unseen test data, with perfect classification of Still, Standing, and Jumping. The primary failure mode, Walking vs Jumping confusion, arises from overlapping periodic frequency features and limited training data coverage. The modular pipeline design enables straightforward extension with additional recordings, features, or activities.